

Walmart Weekly Sales Predictor

A self-contained Docker image serving weekly sales predictions from a trained Random Forest model ($R^2 = 0.9631$ on held-out test data), built on the Walmart Kaggle "Store Sales Forecasting" dataset.

Docker Hub: hub.docker.com/r/ibrahimessam0/walmart-sales-predictor

Run It

```
docker pull ibrahimessam0/walmart-sales-predictor:1.0
docker run -p 8080:8080 ibrahimessam0/walmart-sales-predictor:1.0
```

The API is now available at <http://localhost:8080>.

Get a Prediction

The image serves MLflow's standard model-serving endpoint, `POST /invocations`, using the `dataframe_split` request format.

Linux / Mac / Git Bash

```
curl -X POST http://localhost:8080/invocations \
-H "Content-Type: application/json" \
-d '{
  "dataframe_split": {
    "columns": ["Store", "Dept", "IsHoliday", "Temperature", "Fuel_Price",
               "MarkDown1", "MarkDown2", "MarkDown3", "MarkDown4", "MarkDown5",
               "CPI", "Unemployment", "Size", "Year", "Month", "Week",
               "Quarter", "Day", "Type_B", "Type_C"],
    "data": [[1,1,false,42.31,2.572,0,0,0,0,0,211.096358,8.106,151315,
              2010,2,5,1,5,false,false]]
  }
}'
```

Windows PowerShell

```
$body = @'
{
  "dataframe_split": {
    "columns": ["Store", "Dept", "IsHoliday", "Temperature", "Fuel_Price",
               "MarkDown1", "MarkDown2", "MarkDown3", "MarkDown4", "MarkDown5",
               "CPI", "Unemployment", "Size", "Year", "Month", "Week",
               "Quarter", "Day", "Type_B", "Type_C"],
    "data": [[1,1,false,42.31,2.572,0,0,0,0,0,211.096358,8.106,151315,
              2010,2,5,1,5,false,false]]
  }
}
'@

Invoke-RestMethod -Uri "http://localhost:8080/invocations" -Method Post `
-Body $body -ContentType "application/json"
```

Expected Response

```
{"predictions": [22172.81678364824]}
```

Request Schema

The data array must contain values in exactly this column order:

Column	Type	Description
Store	int	Store number (1-45)
Dept	int	Department number
IsHoliday	bool	Whether the week includes a major holiday
Temperature	float	Average temperature in the region (°F)
Fuel_Price	float	Regional fuel price (USD/gallon)
MarkDown1 - MarkDown5	float	Promotional markdown amounts (use 0 if unknown/not applicable)
CPI	float	Consumer Price Index
Unemployment	float	Regional unemployment rate (%)
Size	int	Store size (sq ft)
Year	int	Calendar year
Month	int	Calendar month (1-12)
Week	int	ISO week number
Quarter	int	Calendar quarter (1-4)
Day	int	Day of month
Type_B	bool	true if store type is "B"
Type_C	bool	true if store type is "C"

Leave Type_B and Type_C both false for store type "A".

You can send multiple rows at once by adding more inner arrays to data — the response will contain one prediction per row, in the same order.

Notes

- This is a snapshot of the model as of the build date below — it does not update automatically. Check Docker Hub for a newer version tag if one is available.
- Model: Random Forest (n_estimators=80, max_depth=15), trained on an 80/20 chronological train/test split of the Walmart dataset.
- Built: July 10, 2026